Power Electronics (Module 2)

15 Numerical Questions & Step-by-Step Solutions

Course: Power Electronics — S5 Electrical & Electronics Engineering — College of Engineering Trivandrum (CET)

References: Daniel W. Hart · Ned Mohan · L. Umanand · Muhammad H. Rashid

Assignment Topic Distribution (15 Questions Total)

No.	Module 2 Topic	Coverage / Parameter Focus	Count
1	1-Phase Fully Controlled Rectifier	R load, RL ripple-free load (THD, PF), RLE load (Inverter mode)	**3**
2	1-Phase Half-Controlled Rectifier	RL continuous load (harmonics, DF), RLE battery charger	**2**
3	Source Inductance Effect	1-Phase full converter commutation overlap (μ, ΔV_o)	**1**
4	3-Phase Controlled Rectifiers	Fully controlled RLE bridge, Inverter mode, Semi-converter bridge	**3**
5	DC-DC Switching Regulators	Buck (Analysis & Design), Boost (Analysis/ESR & Design), Buck-Boost (Analysis & Design)	**6**
		Total Questions Required & Solved:	**15**

Section 1: Single-Phase Fully Controlled Rectifiers (3 Questions)

Question 1: Single-Phase Full Converter with Resistive Load

Textbook Reference: Daniel W. Hart, *Power Electronics*, Example 4-6 / L. Umanand, Section 3.6.

Problem Statement:
A single-phase fully controlled full-wave bridge rectifier is connected to a 120 V (rms), 60 Hz single-phase AC source and supplies a purely resistive load of $R = 20\ \Omega$. The thyristor delay (firing) angle is set to $\alpha = 40°$. Assuming ideal switching devices:

1. Determine the average (DC) output voltage ($V_{o,\mathrm{dc}}$) and average load current ($I_{o,\mathrm{dc}}$).
2. Determine the RMS output voltage ($V_{o,\mathrm{rms}}$) and RMS load current ($I_{o,\mathrm{rms}}$).
3. Calculate the real active power (P) absorbed by the load resistance.
4. Calculate the apparent power (S) supplied by the AC source and the input power factor (PF).

Solution:

- **Given Data:** $V_s = 120$ V (rms), $f = 60$ Hz, $R = 20\ \Omega$, $\alpha = 40° = 0.69813$ rad.
- Peak supply voltage: $V_m = \sqrt{2} \times 120\ \mathrm{V} = 169.706$ V.

1. Average Output Voltage and Average Current:
For a purely resistive load, the current ceases when the input voltage crosses zero at $\omega t = \pi$:

$$V_{o,\mathrm{dc}} = \frac{1}{\pi}\int_{\alpha}^{\pi} V_m \sin(\omega t)\, d(\omega t) = \frac{V_m}{\pi}(1 + \cos\alpha)$$

$$V_{o,\mathrm{dc}} = \frac{169.706}{\pi}(1 + \cos 40°) = 54.019 \times (1 + 0.76604) = 54.019 \times 1.76604 = \mathbf{95.40}\ \mathrm{V}$$

$$I_{o,\mathrm{dc}} = \frac{V_{o,\mathrm{dc}}}{R} = \frac{95.40\ \mathrm{V}}{20\ \Omega} = \mathbf{4.770}\ \mathrm{A}$$

2. RMS Output Voltage and RMS Current:

$$V_{o,\text{rms}} = \sqrt{\frac{1}{\pi}\int_{\alpha}^{\pi} [V_m \sin(\omega t)]^2 \, d(\omega t)} = \frac{V_m}{\sqrt{2}}\sqrt{\frac{\pi - \alpha + \frac{1}{2}\sin(2\alpha)}{\pi}} = V_s\sqrt{\frac{\pi - \alpha + \frac{1}{2}\sin(2\alpha)}{\pi}}$$

Evaluating the angle term: $\pi - \alpha = 3.14159 - 0.69813 = 2.44346$ rad, and $\frac{1}{2}\sin(80^\circ) = \frac{1}{2}(0.98481) = 0.49240$.

$$\text{Radicand} = \frac{2.44346 + 0.49240}{3.14159} = \frac{2.93586}{3.14159} = 0.93451 \implies \sqrt{0.93451} = 0.96670$$

$$V_{o,\text{rms}} = 120 \times 0.96670 = \mathbf{116.00}\ \text{V}, \qquad I_{o,\text{rms}} = \frac{116.00\ \text{V}}{20\ \Omega} = \mathbf{5.800}\ \text{A}$$

3. Real Power Delivered to Load:

$$P = I_{o,\text{rms}}^2 R = (5.800\ \text{A})^2 \times 20\ \Omega = 33.64 \times 20 = \mathbf{672.85}\ \text{W}$$

4. Apparent Power and Power Factor:
For this single-phase circuit, the source RMS current equals the load RMS current ($I_{s,\text{rms}} = I_{o,\text{rms}} = 5.800$ A):

$$S = V_s I_{s,\text{rms}} = 120\ \text{V} \times 5.800\ \text{A} = \mathbf{696.00}\ \text{VA}$$

$$PF = \frac{P}{S} = \frac{672.85\ \text{W}}{696.00\ \text{VA}} = \mathbf{0.9667}\ \text{(lagging)}$$

Summary of Results (Question 1):
$V_{o,\text{dc}} = 95.40$ V | $I_{o,\text{dc}} = 4.770$ A | $V_{o,\text{rms}} = 116.00$ V | $I_{o,\text{rms}} = 5.800$ A | $P = 672.85$ W | $PF = 0.9667$ (lag)

Question 2: Single-Phase Full Converter with RL Load (Continuous, Ripple-Free Current)

Textbook Reference: Daniel W. Hart, Example 4-8 / Ned Mohan, Section 6-3 / L. Umanand, Section 3.4.

Problem Statement:
A single-phase full converter is supplied from a 230 V (rms), 50 Hz AC utility. The converter delivers power to an RL load with $R = 10.0\ \Omega$ and a very large series filter inductance L such that the load current is continuous and virtually ripple-free (I_o = constant). The firing delay angle is $\alpha = 45^\circ$.

1. Compute the average DC output voltage (V_o) and the average DC load current (I_o).
2. Determine the real active power (P) absorbed by the load.
3. Find the RMS value of the AC line current ($I_{s,\text{rms}}$) and the RMS fundamental current ($I_{s1,\text{rms}}$).
4. Calculate the displacement factor (DF), total input power factor (PF), and the Total Harmonic Distortion (THD).

Solution:

- **Given Data:** $V_s = 230$ V, $f = 50$ Hz, $R = 10.0\ \Omega$, $\alpha = 45^\circ$.
- Peak voltage: $V_m = \sqrt{2} \times 230 = 325.27$ V.

1. Average Output Voltage and DC Current:
In continuous conduction, output voltage continues into the negative half-cycle until commutated at $\pi + \alpha$:

$$V_o = \frac{1}{\pi}\int_{\alpha}^{\pi+\alpha} V_m \sin(\omega t)\, d(\omega t) = \frac{2V_m}{\pi}\cos\alpha$$

$$V_o = \frac{2 \times 325.27}{\pi}\cos(45^\circ) = 207.07 \times 0.70711 = \mathbf{146.42}\ \text{V}$$

Because load current is ripple-free, the inductor average voltage drop is zero:

$$I_o = \frac{V_o}{R} = \frac{146.42\ \text{V}}{10.0\ \Omega} = \mathbf{14.642}\ \text{A}$$

2. Real Power Delivered to the Load:
With ripple-free current, $I_{o,\text{rms}} = I_o = 14.642$ A:

$$P = I_o^2 R = (14.642 \text{ A})^2 \times 10.0\ \Omega = \mathbf{2143.9} \text{ W} \quad (\text{also } P = V_o I_o = 146.42 \times 14.642 = 2143.9 \text{ W})$$

3. Input Source Current Analysis:
The AC line current is a square wave of height $\pm I_o$ with conduction duration π in each half-cycle:

$$I_{s,\text{rms}} = I_o = \mathbf{14.642} \text{ A}$$

The fundamental Fourier component of this square wave has RMS magnitude:

$$I_{s1,\text{rms}} = \frac{2\sqrt{2}}{\pi} I_o = 0.90032 \times 14.642 \text{ A} = \mathbf{13.183} \text{ A}$$

4. Displacement Factor, Power Factor, and THD:

- **Displacement Factor (DF):** The fundamental input current lags the voltage by α:

$$DF = \cos\phi_1 = \cos\alpha = \cos(45^\circ) = \mathbf{0.7071}$$

- **Distortion Factor:**

$$\text{Distortion Factor} = \frac{I_{s1,\text{rms}}}{I_{s,\text{rms}}} = \frac{2\sqrt{2}}{\pi} \approx 0.90032$$

- **Total Power Factor (PF):**

$$PF = \left(\frac{I_{s1,\text{rms}}}{I_{s,\text{rms}}}\right)\cos\phi_1 = \frac{2\sqrt{2}}{\pi}\cos(45^\circ) = 0.90032 \times 0.70711 = \mathbf{0.6366} \text{ (lagging)}$$

- **Total Harmonic Distortion (THD):**

$$THD = \sqrt{\left(\frac{I_{s,\text{rms}}}{I_{s1,\text{rms}}}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1} = \sqrt{1.2337 - 1} = \sqrt{0.2337} = \mathbf{48.34\%}$$

Summary of Results (Question 2):
$V_o = 146.42$ V | $I_o = 14.64$ A | $P = 2143.9$ W | $I_{s,\text{rms}} = 14.64$ A | $DF = 0.7071$ | $PF = 0.6366$ | $THD = 48.34\%$

Question 3: Single-Phase Full Converter with RLE Load — Rectifier vs. Inverter Modes

Textbook Reference: Daniel W. Hart, Examples 4-10 & 4-11 / Ned Mohan, Section 6-3-3.

Problem Statement:
A single-phase full converter operates from a 230 V (rms), 50 Hz supply and feeds an active RLE load with $R = 2.0\ \Omega$ and a smoothing inductance large enough to keep the DC current ripple-free.

1. **Rectifier Mode ($\alpha < 90^\circ$):** The DC source is a battery being charged with EMF $E = +60.0$ V. If the charging current is regulated to $I_o = 15.0$ A, find the required firing angle α_{rect}, the power supplied by the AC grid, the power delivered to the battery, and the resistive dissipation.
2. **Inverter Mode ($\alpha > 90^\circ$):** The battery terminals are reversed ($E = -60.0$ V) so that it acts as a DC source delivering regenerative power back into the AC grid. For the same current $I_o = 15.0$ A, determine the inverter firing angle α_{inv}, the power generated by the DC battery, and the net regenerative power returned to the AC grid.

Solution:

- Maximum average voltage: $V_{do} = \frac{2V_m}{\pi} = \frac{2\times\sqrt{2}\times230}{\pi} = 207.07$ V.

1. Rectifier Mode ($\alpha < 90°$):
The DC output voltage required to maintain $I_o = 15.0$ A is:

$$V_o = E + I_o R = 60.0 \text{ V} + (15.0 \text{ A} \times 2.0 \ \Omega) = 60.0 + 30.0 = \mathbf{90.0} \text{ V}$$

Setting $V_o = V_{do} \cos\alpha$:

$$\cos\alpha_{\text{rect}} = \frac{V_o}{V_{do}} = \frac{90.0 \text{ V}}{207.07 \text{ V}} = 0.43463 \implies \alpha_{\text{rect}} = \cos^{-1}(0.43463) = \mathbf{64.24}°$$

Power components:

- Power absorbed by battery: $P_E = E \times I_o = 60.0 \text{ V} \times 15.0 \text{ A} = \mathbf{900.0}$ W.
- Power dissipated in resistance: $P_R = I_o^2 R = (15)^2 \times 2.0 = \mathbf{450.0}$ W.
- Total power supplied by AC source: $P_{\text{ac}} = V_o I_o = 90.0 \times 15.0 = \mathbf{1350.0}$ W $\quad (= P_E + P_R)$.

2. Inverter Mode ($\alpha > 90°$):
Since thyristors conduct in only one direction, reversing the power flow requires reversing the DC voltage polarity:

$$V_o = -E + I_o R = -60.0 \text{ V} + (15.0 \text{ A} \times 2.0 \ \Omega) = -60.0 + 30.0 = \mathbf{-30.0} \text{ V}$$

The negative polarity indicates the bridge acts as an inverter:

$$\cos\alpha_{\text{inv}} = \frac{V_o}{V_{do}} = \frac{-30.0 \text{ V}}{207.07 \text{ V}} = -0.14488 \implies \alpha_{\text{inv}} = \cos^{-1}(-0.14488) = \mathbf{98.33}°$$

Power components:

- Power supplied by DC battery source: $P_{\text{dc, gen}} = |E| \times I_o = 60.0 \text{ V} \times 15.0 \text{ A} = \mathbf{900.0}$ W.
- Power lost in resistance R: $P_R = I_o^2 R = (15)^2 \times 2.0 = \mathbf{450.0}$ W.
- Net power regenerated back to AC utility grid: $P_{\text{grid}} = |V_o| I_o = 30.0 \times 15.0 = \mathbf{450.0}$ W $\quad (= P_{\text{dc, gen}} - P_R)$.

Summary of Results (Question 3):
Rectifier Mode: $V_o = +90.0$ V, $\alpha = 64.24°$, $P_{\text{ac}} = 1350$ W, $P_{\text{batt}} = 900$ W, $P_R = 450$ W
Inverter Mode: $V_o = -30.0$ V, $\alpha = 98.33°$, $P_{\text{dc, gen}} = 900$ W, $P_R = 450$ W, $P_{\text{grid}} = 450$ W

Section 2: Single-Phase Half-Controlled Rectifiers (2 Questions)

Question 4: Single-Phase Semi-converter with RL Load — Harmonics & Power Factor

Textbook Reference: Muhammad H. Rashid, Chapter 11 / L. Umanand, Section 3.6.

Problem Statement:
A single-phase half-controlled (semi-converter) bridge rectifier supplies an RL load with $R = 10.0\ \Omega$ and a large smoothing reactor ensuring continuous, ripple-free current. The AC supply is 230 V (rms), 50 Hz. The thyristor firing delay angle is $\alpha = 60°$.

1. Calculate the average output voltage ($V_{o,\text{dc}}$) and DC load current (I_o).
2. Calculate the RMS output voltage ($V_{o,\text{rms}}$).
3. Determine the RMS input line current ($I_{s,\text{rms}}$) and the fundamental RMS component ($I_{s1,\text{rms}}$).
4. Calculate the displacement factor (DF), distortion factor, total input power factor (PF), and THD.

Solution:

- **Given Data:** $V_s = 230$ V, $V_m = \sqrt{2} \times 230 = 325.27$ V, $R = 10.0\ \Omega$, $\alpha = 60° = \pi/3$ rad.

1. Average Output Voltage and Current:
Due to the freewheeling path, the voltage cannot be negative and is non-zero from α to π:

$$V_{o,\text{dc}} = \frac{1}{\pi}\int_{\alpha}^{\pi} V_m \sin(\omega t)\, d(\omega t) = \frac{V_m}{\pi}(1+\cos\alpha)$$

$$V_{o,\text{dc}} = \frac{325.27}{\pi}(1+\cos 60^\circ) = \frac{325.27 \times 1.5}{3.14159} = \mathbf{155.30}\ \text{V}$$

$$I_o = \frac{V_{o,\text{dc}}}{R} = \frac{155.30\ \text{V}}{10.0\ \Omega} = \mathbf{15.530}\ \text{A}$$

2. RMS Output Voltage:

$$V_{o,\text{rms}} = V_s\sqrt{\frac{\pi - \alpha + \frac{1}{2}\sin(2\alpha)}{2\pi}} \quad \text{where } \alpha = \frac{\pi}{3}\ \text{rad},\ 2\alpha = 120^\circ$$

$$\text{Numerator} = \left(\pi - \frac{\pi}{3}\right) + \frac{1}{2}\sin(120^\circ) = \frac{2\pi}{3} + \frac{\sqrt{3}}{4} = 2.0944 + 0.4330 = 2.5274$$

$$V_{o,\text{rms}} = 230 \times \sqrt{\frac{2.5274}{2\pi}} = 230 \times \sqrt{0.40225} = 230 \times 0.8969 = \mathbf{206.30}\ \text{V}$$

3. Input Source Current Analysis:
Current is drawn from the AC source only when an SCR-diode pair conducts (duration $\pi - \alpha$ per half cycle):

$$I_{s,\text{rms}} = I_o\sqrt{\frac{\pi-\alpha}{\pi}} = 15.530 \times \sqrt{\frac{180^\circ - 60^\circ}{180^\circ}} = 15.530 \times \sqrt{\frac{2}{3}} = 15.530 \times 0.81650 = \mathbf{12.680}\ \text{A}$$

Fundamental RMS input current for a semi-converter is:

$$I_{s1,\text{rms}} = \frac{2\sqrt{2}}{\pi} I_o \cos\left(\frac{\alpha}{2}\right) = \frac{2\sqrt{2} \times 15.530}{\pi}\cos(30^\circ) = 13.982 \times 0.86603 = \mathbf{12.109}\ \text{A}$$

4. Performance Metrics:

- **Displacement Factor:** $DF = \cos\left(\frac{\alpha}{2}\right) = \cos(30^\circ) = \mathbf{0.8660}$.
- **Distortion Factor:** $\frac{I_{s1,\text{rms}}}{I_{s,\text{rms}}} = \frac{12.109}{12.680} = \mathbf{0.9550}$.
- **Power Factor:** PF = Distortion Factor $\times\ DF = 0.9550 \times 0.8660 = \mathbf{0.8270}$ (lagging).
- **THD:**

$$THD = \sqrt{\left(\frac{I_{s,\text{rms}}}{I_{s1,\text{rms}}}\right)^2 - 1} = \sqrt{\left(\frac{12.680}{12.109}\right)^2 - 1} = \sqrt{(1.0471)^2 - 1} = \mathbf{31.08}\%$$

Summary of Results (Question 4):
$V_{o,\text{dc}} = 155.30$ V | $I_o = 15.53$ A | $V_{o,\text{rms}} = 206.30$ V | $I_{s,\text{rms}} = 12.68$ A | $DF = 0.8660$ | $PF = 0.8270$ | $THD = 31.08\%$

Question 5: Single-Phase Semi-converter with RLE Load (Battery Charger)

Textbook Reference: Muhammad H. Rashid, Chapter 11 / Ned Mohan, Chapter 6.
Problem Statement:
A single-phase semi-converter is employed to charge a battery rated at $E = 110.0$ V with an internal resistance of $R = 2.50\ \Omega$. A large series inductor is connected to maintain a ripple-free DC current. The converter is connected to a 230 V (rms), 50 Hz supply. The thyristors are fired at $\alpha = 45^\circ$.

1. Calculate the average DC output voltage (V_o) of the converter.
2. Determine the average charging current (I_o).
3. Calculate the power delivered to the battery (P_E), power dissipated in resistance (P_R), and total power

(P_{total}).

4. Determine the RMS value of the input current ($I_{s,\text{rms}}$) and the input power factor (PF).

Solution:

- **Given Data:** $V_s = 230$ V, $V_m = \sqrt{2} \times 230 = 325.27$ V, $E = 110.0$ V, $R = 2.50\ \Omega$, $\alpha = 45°$.

1. Average Output Voltage:

$$V_o = \frac{V_m}{\pi}(1 + \cos\alpha) = \frac{325.27}{\pi}(1 + \cos 45°) = 103.537 \times 1.70711 = \mathbf{176.75}\text{ V}$$

2. Average Charging Current:
Applying KVL in the DC loop (inductor voltage is zero in steady state):

$$I_o = \frac{V_o - E}{R} = \frac{176.75\text{ V} - 110.0\text{ V}}{2.50\ \Omega} = \frac{66.75}{2.50} = \mathbf{26.70}\text{ A}$$

3. Power Calculations:

- Power delivered to battery: $P_E = E \times I_o = 110.0\text{ V} \times 26.70\text{ A} = \mathbf{2937.0}$ W.
- Power dissipated in internal resistance: $P_R = I_o^2 R = (26.70)^2 \times 2.50 = 712.89 \times 2.50 = \mathbf{1782.2}$ W.
- Total power output: $P_{\text{total}} = P_E + P_R = 2937.0 + 1782.2 = \mathbf{4719.2}$ W $\quad (= V_o I_o = 176.75 \times 26.70)$.

4. RMS Input Current and Power Factor:

$$I_{s,\text{rms}} = I_o\sqrt{\frac{\pi - \alpha}{\pi}} = 26.70 \times \sqrt{\frac{180° - 45°}{180°}} = 26.70 \times \sqrt{0.75} = 26.70 \times 0.86603 = \mathbf{23.123}\text{ A}$$

$$S = V_s I_{s,\text{rms}} = 230\text{ V} \times 23.123\text{ A} = 5318.3\text{ VA}$$

$$PF = \frac{P_{\text{total}}}{S} = \frac{4719.2\text{ W}}{5318.3\text{ VA}} = \mathbf{0.8874}\text{ (lagging)}$$

Summary of Results (Question 5):
$V_o = 176.75$ V | $I_o = 26.70$ A | $P_E = 2937.0$ W | $P_R = 1782.2$ W | $P_{\text{total}} = 4719.2$ W | $I_{s,\text{rms}} = 23.12$ A | $PF = 0.8874$

Section 3: Effect of Source Inductance (1 Question)

Question 6: Commutation Overlap in Single-Phase Full Converter

Textbook Reference: Daniel W. Hart, Section 4.7 / Ned Mohan, Section 6-3-4 / L. Umanand.

Problem Statement:
A single-phase fully controlled bridge converter is fed from a 230 V (rms), 50 Hz supply. Due to transformer leakage, the circuit possesses an effective source inductance of $L_s = 2.0$ mH. The converter supplies a continuous, ripple-free load current of $I_o = 15.0$ A. The firing delay angle is $\alpha = 30°$.

1. Determine the commutation (overlap) angle μ in degrees.
2. Calculate the average DC voltage drop (ΔV_o) caused by the commutation overlap.
3. Find the actual average DC output voltage (V_o).
4. Determine the ideal output voltage ($V_{o,\text{ideal}}$ for $L_s = 0$) and the percentage voltage regulation due to source inductance.

Solution:

- **Given Data:** $V_s = 230$ V, $f = 50$ Hz $\implies$ $\omega = 2\pi(50) = 314.159$ rad/s, $L_s = 2.0 \times 10^{-3}$ H, $I_o = 15.0$ A, $\alpha = 30°$.
- Peak voltage: $V_m = \sqrt{2} \times 230 = 325.27$ V.
- Source reactance: $X_s = \omega L_s = 314.159 \times 0.002 = 0.62832\ \Omega$.

1. Commutation / Overlap Angle μ:
During commutation, current through L_s changes from $-I_o$ to $+I_o$:

$$\cos(\alpha + \mu) = \cos\alpha - \frac{2\omega L_s I_o}{V_m}$$

Calculate the reactance reduction term:

$$\frac{2\omega L_s I_o}{V_m} = \frac{2 \times 0.62832 \times 15.0}{325.27} = \frac{18.8496}{325.27} = 0.05795$$

Since $\alpha = 30° \implies \cos(30°) = 0.86603$:

$$\cos(\alpha + \mu) = 0.86603 - 0.05795 = 0.80808$$

$$\alpha + \mu = \cos^{-1}(0.80808) = 36.09° \implies \mu = 36.09° - 30.00° = \mathbf{6.09°}$$

2. Average Voltage Drop Due to Overlap:
During overlap, all four SCRs conduct, shorting the bridge and making terminal voltage zero:

$$\Delta V_o = \frac{2\omega L_s I_o}{\pi} = \frac{2(2\pi f) L_s I_o}{\pi} = 4 f L_s I_o$$

$$\Delta V_o = 4 \times 50\ \text{Hz} \times (2.0 \times 10^{-3}\ \text{H}) \times 15.0\ \text{A} = \mathbf{6.00}\ \text{V}$$

3. Actual Average DC Output Voltage:
The ideal output voltage without overlap is:

$$V_{o,\text{ideal}} = \frac{2V_m}{\pi}\cos\alpha = \frac{2 \times 325.27}{\pi}\cos(30°) = 207.07 \times 0.86603 = \mathbf{179.33}\ \text{V}$$

The actual output voltage is:

$$V_o = V_{o,\text{ideal}} - \Delta V_o = 179.33\ \text{V} - 6.00\ \text{V} = \mathbf{173.33}\ \text{V}$$

4. Percentage Voltage Drop:

$$\%\ \text{Reduction} = \frac{\Delta V_o}{V_{o,\text{ideal}}} \times 100 = \frac{6.00}{179.33} \times 100 = \mathbf{3.35}\%$$

Summary of Results (Question 6):
Overlap Angle $\mu = 6.09°$ | $\Delta V_o = 6.00$ V | $V_{o,\text{actual}} = 173.33$ V | $V_{o,\text{ideal}} = 179.33$ V | % Drop = 3.35%

Section 4: Three-Phase Controlled Rectifiers (3 Questions)

Question 7: Three-Phase Fully Controlled Converter with RLE Load

Textbook Reference: Daniel W. Hart, Example 4-13 / L. Umanand, Section 3.6 / M. H. Rashid, Section 12.2.

Problem Statement:
A 3-phase, 6-pulse fully controlled bridge converter is fed from a 415 V (rms, line-to-line), 50 Hz supply. It feeds an RLE load with $R = 4.0\ \Omega$, back EMF $E = 220.0$ V, and a large inductor that keeps the load current continuous and ripple-free. The thyristor delay angle is $\alpha = 45°$.

1. State the output voltage equation and compute the average output voltage (V_o).
2. Determine the average load current (I_o).
3. Compute the power absorbed by the EMF (P_E), power lost in resistance (P_R), and total delivered power (P_{total}).
4. Calculate the RMS source line current ($I_{s,\text{rms}}$) and the input power factor (PF).

Solution:

- **Given Data:** $V_{LL} = 415$ V, $f = 50$ Hz, $R = 4.0\ \Omega$, $E = 220.0$ V, $\alpha = 45°$.
- Peak line-to-line voltage: $V_{m,LL} = \sqrt{2} \times 415 = 586.899$ V.

1. Output Voltage Equation and Value:
Integrating over a 60° conduction interval:

$$V_o = \frac{3}{\pi} \int_{\pi/3+\alpha}^{2\pi/3+\alpha} V_{m,LL} \sin(\omega t)\, d(\omega t) = \frac{3V_{m,LL}}{\pi} \cos\alpha = \frac{3\sqrt{2}V_{LL}}{\pi} \cos\alpha$$

$$V_o = \frac{3 \times 586.899}{\pi} \cos(45°) = 560.45 \times 0.70711 = \mathbf{396.30}\ \text{V}$$

2. Average Load Current:

$$I_o = \frac{V_o - E}{R} = \frac{396.30\ \text{V} - 220.0\ \text{V}}{4.0\ \Omega} = \frac{176.30}{4.0} = \mathbf{44.075}\ \text{A}$$

3. Power Calculations:

- Power absorbed by battery: $P_E = E \times I_o = 220.0\ \text{V} \times 44.075\ \text{A} = \mathbf{9696.5}$ W.
- Resistive loss: $P_R = I_o^2 R = (44.075)^2 \times 4.0 = \mathbf{7770.4}$ W.
- Total power output: $P_{\text{total}} = V_o I_o = 396.30 \times 44.075 = \mathbf{17466.9}$ W $\quad (= P_E + P_R)$.

4. RMS Line Current and Input Power Factor:
Each line conducts pulses of width 120° in positive and negative half-cycles:

$$I_{s,\text{rms}} = I_o \sqrt{\frac{2}{3}} = 44.075 \times 0.81650 = \mathbf{35.987}\ \text{A}$$

$$S = \sqrt{3} V_{LL} I_{s,\text{rms}} = \sqrt{3} \times 415 \times 35.987 = \mathbf{25867.7}\ \text{VA}$$

$$PF = \frac{P_{\text{total}}}{S} = \frac{17466.9\ \text{W}}{25867.7\ \text{VA}} = \mathbf{0.6752}\ \text{(lagging)} \quad \left(\text{Check: } \frac{3}{\pi}\cos(45°) = 0.6752\right)$$

Summary of Results (Question 7):
$V_o = 396.30$ V | $I_o = 44.08$ A | $P_{\text{total}} = 17.47$ kW | $I_{s,\text{rms}} = 35.99$ A | $PF = 0.6752$ (lag)

Question 8: Three-Phase Full Converter Operating as a Line-Commutated Inverter

Textbook Reference: Daniel W. Hart, Example 4-14 / Ned Mohan, Section 6-4.

Problem Statement:
A 3-phase six-pulse converter is operated as a line-commutated inverter to regenerate energy from a DC source ($E_{\text{dc}} = 320.0$ V) into a 3-phase, 415 V (rms, line-to-line), 50 Hz AC power system. The internal loop resistance is $R = 2.0\ \Omega$ with a large smoothing inductance. The thyristor delay angle is $\alpha = 120°$ ($\alpha > 90°$).

1. Calculate the average DC output voltage (V_o) of the converter.
2. Determine the DC current (I_o) fed into the inverter.

3. Calculate the power delivered by the DC source (P_{dc}), the power loss in resistance (P_{loss}), and the net power delivered to the AC grid (P_{grid}).

Solution:

- **Given Data:** $V_{LL} = 415$ V, $E_{\text{dc}} = 320.0$ V, $R = 2.0\ \Omega$, $\alpha = 120°$.

1. Converter DC Output Voltage:

$$V_o = \frac{3\sqrt{2}V_{LL}}{\pi}\cos\alpha = 560.45 \times \cos(120°) = 560.45 \times (-0.5) = \mathbf{-280.22}\ \text{V}$$

The negative polarity confirms inversion mode.

2. Inverter DC Current:

The DC source voltage opposes the bridge counter-voltage:

$$I_o = \frac{E_{\text{dc}} - |V_o|}{R} = \frac{320.0\ \text{V} - 280.22\ \text{V}}{2.0\ \Omega} = \frac{39.78}{2.0} = \mathbf{19.89}\ \text{A}$$

3. Power Calculations:

- Power supplied by DC source: $P_{\text{dc}} = E_{\text{dc}} \times I_o = 320.0\ \text{V} \times 19.89\ \text{A} = \mathbf{6364.8}\ \text{W}$.
- Resistive loss: $P_{\text{loss}} = I_o^2 R = (19.89)^2 \times 2.0 = \mathbf{791.2}\ \text{W}$.
- Net power regenerated into AC grid: $P_{\text{grid}} = |V_o| I_o = 280.22\ \text{V} \times 19.89\ \text{A} = \mathbf{5573.6}\ \text{W} \quad (= P_{\text{dc}} - P_{\text{loss}})$.

Summary of Results (Question 8):
$V_o = -280.22$ V | $I_o = 19.89$ A | $P_{\text{dc}} = 6364.8$ W | $P_{\text{loss}} = 791.2$ W | $P_{\text{grid}} = 5573.6$ W

Question 9: Three-Phase Half-Controlled Bridge Converter (Semi-converter)

Textbook Reference: L. Umanand, Eq. 3.74 / Muhammad H. Rashid, Section 12.2.5.

Problem Statement:
A 3-phase half-controlled bridge converter (3 SCRs and 3 uncontrolled diodes) is supplied from a 415 V (line-to-line, rms), 50 Hz supply. It feeds an RLE load with $R = 6.0\ \Omega$, $E = 150.0$ V, and large inductance. The firing angle is $\alpha = 60°$.

1. State the average output voltage equation and compute V_o.
2. Find the average load current (I_o) and the total power delivered to the load (P_{total}).
3. Compare the average output voltage with that of a fully controlled 3-phase bridge at $\alpha = 60°$.

Solution:

- **Given Data:** $V_{LL} = 415$ V, $R = 6.0\ \Omega$, $E = 150.0$ V, $\alpha = 60°$.

1. Output Voltage Equation and Value:

For a 3-phase semi-converter:

$$V_o = \frac{3\sqrt{2}V_{LL}}{2\pi}(1 + \cos\alpha) = \frac{560.45}{2}(1 + \cos 60°) = 280.224 \times (1 + 0.5) = \mathbf{420.34}\ \text{V}$$

2. Average Load Current and Total Power:

$$I_o = \frac{V_o - E}{R} = \frac{420.34\ \text{V} - 150.0\ \text{V}}{6.0\ \Omega} = \frac{270.34}{6.0} = \mathbf{45.06}\ \text{A}$$

$$P_{\text{total}} = V_o I_o = 420.34\ \text{V} \times 45.06\ \text{A} = \mathbf{18940.5}\ \text{W}$$

- Power to EMF: $P_E = 150.0 \times 45.06 = 6759.0$ W.

- Resistive loss: $P_R = (45.06)^2 \times 6.0 = 12182.4$ W.
- Check: $6759.0 + 12182.4 = 18941.4$ W $\approx P_{\text{total}}$.

3. Comparison with Fully Controlled Bridge at $\alpha = 60°$:

$$V_{o,\text{full}} = \frac{3\sqrt{2}V_{LL}}{\pi}\cos(60°) = 560.45 \times 0.5 = \mathbf{280.22}\text{ V}$$

Observation: The semi-converter produces a higher DC output voltage (420.34 V vs 280.22 V) because the free-wheeling diodes prevent the line-to-line output voltage from swinging negative.

Summary of Results (Question 9):
$V_{o,\text{semi}} = 420.34$ V | $I_o = 45.06$ A | $P_{\text{total}} = 18.94$ kW | $V_{o,\text{full}} = 280.22$ V

Section 5: DC-DC Switching Regulators (6 Questions)

Question 10: Buck Converter Analysis in Continuous Conduction Mode (CCM)

Textbook Reference: Daniel W. Hart, Example 6-1 / Ned Mohan, Section 7-3.

Problem Statement:
A step-down Buck converter operates with the following parameters: $V_s = 50.0$ V, duty ratio $D = 0.40$, filter inductance $L = 400.0$ μH, filter capacitance $C = 100.0$ μF, switching frequency $f = 20.0$ kHz, and load resistance $R = 20.0$ Ω.

1. Determine the output voltage (V_o) and load current (I_o).
2. Calculate the peak-to-peak inductor ripple current (Δi_L) and find $I_{\max}$ and $I_{\min}$.
3. Verify whether the converter operates in Continuous Conduction Mode (CCM).
4. Calculate the peak-to-peak output voltage ripple (ΔV_o) and percentage ripple.

Solution:

- **Given Data:** $V_s = 50.0$ V, $D = 0.40$, $L = 400$ μH, $C = 100$ μF, $f = 20$ kHz $\implies T = 50$ μs, $R = 20$ Ω.

1. Output Voltage and Current:

$$V_o = DV_s = 0.40 \times 50.0\text{ V} = \mathbf{20.0}\text{ V}, \qquad I_o = I_L = \frac{V_o}{R} = \frac{20.0\text{ V}}{20.0\ \Omega} = \mathbf{1.00}\text{ A}$$

2. Inductor Ripple Current:
During switch ON-interval ($DT = 20$ μs):

$$\Delta i_L = \frac{(V_s - V_o)DT}{L} = \frac{(50-20)\times 0.40 \times (50\times 10^{-6})}{400\times 10^{-6}} = \frac{30\times 20}{400} = \mathbf{1.50}\text{ A}$$

$$I_{\max} = I_L + \frac{\Delta i_L}{2} = 1.00 + 0.75 = \mathbf{1.75}\text{ A}, \qquad I_{\min} = I_L - \frac{\Delta i_L}{2} = 1.00 - 0.75 = \mathbf{0.25}\text{ A}$$

3. CCM Verification:
Since $I_{\min} = 0.25$ A > 0, current never falls to zero. **CCM operation is verified.**

4. Output Voltage Ripple:

$$\Delta V_o = \frac{\Delta i_L}{8Cf} = \frac{1.50\text{ A}}{8\times(100\times 10^{-6})\times(20\times 10^3)} = \frac{1.50}{16.0} = \mathbf{0.09375}\text{ V} = \mathbf{93.75}\text{ mV}$$

$$\%\text{ Ripple} = \frac{\Delta V_o}{V_o}\times 100 = \frac{0.09375}{20.0}\times 100 = \mathbf{0.469}\%$$

Summary of Results (Question 10):
$V_o = 20.0$ V | $I_o = 1.00$ A | $\Delta i_L = 1.50$ A | $I_{\max} = 1.75$ A | $I_{\min} = 0.25$ A | $\Delta V_o = 93.75$ mV (0.469%)

Question 11: Buck Converter Design — Sizing Filter and Device Ratings

Textbook Reference: Daniel W. Hart, Example 6-2 / Ned Mohan, Section 7-3.

Problem Statement:
Design a Buck converter to deliver $V_o = 18.0$ V across $R = 10.0\ \Omega$ from a DC supply of $V_s = 48.0$ V. The specifications are: $f = 40.0$ kHz, voltage ripple $\Delta V_o/V_o \leq 0.5\%$, and CCM operation.

1. Determine the duty ratio D.
2. Find the critical inductance ($L_{\min}$) for CCM, and choose a practical inductor L with a 60% safety margin.
3. Sizing the output filter capacitance C.
4. Specify the peak voltage and peak current ratings for the active switch (MOSFET) and diode.

Solution:

- **Given Data:** $V_s = 48.0$ V, $V_o = 18.0$ V, $R = 10.0\ \Omega$, $f = 40$ kHz, $\Delta V_o/V_o \leq 0.005$.

1. Duty Ratio and Load Current:

$$D = \frac{V_o}{V_s} = \frac{18.0}{48.0} = \mathbf{0.375}, \qquad I_o = \frac{V_o}{R} = \frac{18.0}{10.0} = \mathbf{1.80}\ \text{A}$$

2. Minimum Inductance and Inductor Selection:

$$L_{\min} = \frac{(1-D)R}{2f} = \frac{(1-0.375)\times 10.0}{2\times 40000} = \frac{6.25}{80000} = \mathbf{78.125}\ \mu\text{H}$$

Selecting L with a 60% design margin ($L = 1.6 \times L_{\min}$):

$$L = 1.6 \times 78.125\ \mu\text{H} = \mathbf{125.0}\ \mu\text{H}$$

With $L = 125.0\ \mu$H:

$$\Delta i_L = \frac{(V_s - V_o)D}{Lf} = \frac{(48-18)\times 0.375}{(125\times 10^{-6})\times(40\times 10^3)} = \frac{11.25}{5.0} = 2.25\ \text{A}$$

$$I_{\text{peak}} = I_o + \frac{\Delta i_L}{2} = 1.80 + 1.125 = \mathbf{2.925}\ \text{A}$$

3. Filter Capacitance Sizing:

$$C \geq \frac{1-D}{8Lf^2(\Delta V_o/V_o)} = \frac{1-0.375}{8\times(125\times 10^{-6})\times(40\times 10^3)^2\times 0.005} = \frac{0.625}{8000} = \mathbf{78.125}\ \mu\text{F}$$

Choose standard value: 100 μ**F**.

4. Power Device Ratings:

- **MOSFET Switch:** Peak voltage $V_{\text{switch, peak}} = V_s = \mathbf{48.0}$ V (select ≥ 100 V rated). Peak current: $I_{\text{switch, peak}} = I_{\text{peak}} = \mathbf{2.925}$ A. Average current: $I_{\text{avg}} = DI_o = 0.375 \times 1.80 = \mathbf{0.675}$ A.
- **Freewheeling Diode:** Peak reverse voltage $V_{\text{diode, peak}} = V_s = \mathbf{48.0}$ V. Peak forward current: **2.925** A. Average forward current: $I_{\text{diode, avg}} = (1-D)I_o = 0.625 \times 1.80 = \mathbf{1.125}$ A.

Summary of Results (Question 11):
$D = 0.375$ | $L_{\min} = 78.13\ \mu$H | Design $L = 125\ \mu$H | $C \geq 78.13\ \mu$F (use 100 μF) | $V_{\text{device}} \geq 48$ V, $I_{\text{peak}} = 2.925$ A

Question 12: Boost Converter Analysis and Effect of Inductor Resistance

Textbook Reference: Daniel W. Hart, Examples 6-4 & 6-5 / Ned Mohan, Section 7-4.

Problem Statement:
A Boost converter operates from $V_s = 12.0$ V and delivers $V_o = 30.0$ V to $R = 50.0\ \Omega$ at $f = 25.0$ kHz. Filter components are $L = 120.0\ \mu$H and $C = 100.0\ \mu$F.

1. Assuming ideal components, calculate the duty ratio D, average inductor current (I_L), ripple current (Δi_L), peak current ($I_{\max}$), and output ripple voltage (ΔV_o).
2. If the inductor has an internal winding resistance of $r_L = 0.20\ \Omega$, determine the actual output voltage and converter efficiency η.

Solution:

- **Given Data:** $V_s = 12.0$ V, $V_o = 30.0$ V, $R = 50.0\ \Omega$, $f = 25$ kHz $\implies T = 40\ \mu$s, $L = 120\ \mu$H, $C = 100\ \mu$F.

1. Ideal Boost Converter Analysis:

$$1 - D = \frac{V_s}{V_o} = \frac{12.0}{30.0} = 0.40 \implies \mathbf{D = 0.60}$$

$$I_o = \frac{V_o}{R} = \frac{30.0\text{ V}}{50.0\ \Omega} = 0.60\text{ A}, \qquad I_L = \frac{I_o}{1-D} = \frac{0.60}{0.40} = \mathbf{1.50}\text{ A}$$

$$\Delta i_L = \frac{V_s D T}{L} = \frac{12.0 \times 0.60 \times (40 \times 10^{-6})}{120 \times 10^{-6}} = \mathbf{2.40}\text{ A}$$

$$I_{\max} = I_L + \frac{\Delta i_L}{2} = 1.50 + 1.20 = \mathbf{2.70}\text{ A}, \qquad I_{\min} = 1.50 - 1.20 = \mathbf{0.30}\text{ A (CCM confirmed)}$$

$$\Delta V_o = \frac{I_o D}{C f} = \frac{0.60 \times 0.60}{(100 \times 10^{-6}) \times (25 \times 10^3)} = \frac{0.36}{2.50} = \mathbf{0.144}\text{ V} = \mathbf{144}\text{ mV} \quad (0.48\%)$$

2. Effect of Inductor Resistance $r_L = 0.20\ \Omega$:
Accounting for inductor series resistance:

$$V_{o,\text{actual}} = \left(\frac{V_s}{1-D}\right)\left[\frac{1}{1 + \frac{r_L}{(1-D)^2 R}}\right]$$

$$\frac{r_L}{(1-D)^2 R} = \frac{0.20}{(0.40)^2 \times 50.0} = \frac{0.20}{0.16 \times 50.0} = \frac{0.20}{8.0} = 0.025$$

$$V_{o,\text{actual}} = 30.0 \times \left(\frac{1}{1+0.025}\right) = \frac{30.0}{1.025} = \mathbf{29.27}\text{ V}$$

$$\eta = \frac{1}{1 + \frac{r_L}{(1-D)^2 R}} = \frac{1}{1.025} = \mathbf{97.56}\%$$

Summary of Results (Question 12):
Ideal: $D = 0.60$ | $I_L = 1.50$ A | $\Delta i_L = 2.40$ A | $I_{\max} = 2.70$ A | $\Delta V_o = 144$ mV
With $r_L = 0.2\ \Omega$: $V_{o,\text{actual}} = 29.27$ V | Efficiency $\eta = 97.56\%$

Question 13: Boost Converter Design — Filter Sizing and Power Device Selection

Textbook Reference: Daniel W. Hart, Example 6-4 / Ned Mohan, Section 7-4.

Problem Statement:
Design a Boost converter to step up $V_s = 24.0$ V to $V_o = 48.0$ V supplying $P_o = 120.0$ W ($R = 19.2\ \Omega$) at $f = 50.0$ kHz. The design criteria are: $\Delta V_o / V_o \leq 1.0\%$, and inductor ripple $\Delta i_L \leq 30\%$ of I_L.

1. Determine the duty cycle D and average inductor current (I_L).
2. Determine the critical boundary inductance ($L_{\min}$) and calculate the required design inductance (L).

3. Calculate the required filter capacitance (C).
4. Specify the voltage and current ratings for the active switch and fast diode.

Solution:

- **Given Data:** $V_s = 24.0$ V, $V_o = 48.0$ V, $P_o = 120$ W $\implies I_o = \frac{120}{48} = 2.50$ A, $R = \frac{48}{2.5} = 19.2\ \Omega$, $f = 50$ kHz.

1. Duty Cycle and Inductor Current:

$$1 - D = \frac{V_s}{V_o} = \frac{24.0}{48.0} = 0.50 \implies \mathbf{D = 0.50}, \qquad I_L = \frac{I_o}{1-D} = \frac{2.50}{0.50} = \mathbf{5.00}\ \text{A}$$

2. Inductor Sizing:
Critical inductance at CCM boundary:

$$L_{\min} = \frac{D(1-D)^2 R}{2f} = \frac{0.50 \times (0.50)^2 \times 19.20}{2 \times 50000} = \frac{2.40}{100000} = \mathbf{24.0}\ \mu\text{H}$$

Design for $\Delta i_L \leq 30\% \times I_L = 0.30 \times 5.0$ A $= 1.50$ A:

$$L = \frac{V_s D}{f \Delta i_L} = \frac{24.0 \times 0.50}{(50 \times 10^3) \times 1.50} = \frac{12.0}{75000} = \mathbf{160.0}\ \mu\text{H} \quad (\gg L_{\min})$$

3. Filter Capacitance Sizing:

$$C \geq \frac{D}{Rf(\Delta V_o / V_o)} = \frac{0.50}{19.20 \times (50 \times 10^3) \times 0.01} = \frac{0.50}{9600} = \mathbf{52.08}\ \mu\text{F}$$

Select standard value: 68 μ**F** or 100 μ**F**.
4. Device Selection / Ratings:

- **MOSFET Switch:** Peak voltage stress: $V_{\text{switch, peak}} = V_o = \mathbf{48.0}$ V (select ≥ 100 V). Peak current: $I_{\text{peak}} = I_L + \frac{\Delta i_L}{2} = 5.0 + 0.75 = \mathbf{5.75}$ A. RMS current: $I_{\text{rms}} \approx I_L\sqrt{D} = 5.0 \times \sqrt{0.50} = \mathbf{3.536}$ A.
- **Diode:** Peak reverse voltage: $V_{\text{diode, peak}} = V_o = \mathbf{48.0}$ V. Peak forward current: **5.75** A. Average forward current: $I_{\text{diode, avg}} = I_o = \mathbf{2.50}$ A.

Summary of Results (Question 13):
$D = 0.50$ | $L_{\min} = 24.0\ \mu$H | Designed $L = 160.0\ \mu$H | $C \geq 52.08\ \mu$F | $V_{\text{peak}} = 48$ V, $I_{\text{peak}} = 5.75$ A

Question 14: Buck-Boost Converter Analysis in Continuous Conduction Mode

Textbook Reference: Daniel W. Hart, Example 6-6 / Ned Mohan, Section 7-5.

Problem Statement:
An inverting Buck-Boost DC-DC converter has the following parameters: $V_s = 24.0$ V, duty ratio $D = 0.40$, switching frequency $f = 100.0$ kHz, $L = 20.0\ \mu$H, $C = 80.0\ \mu$F, and load resistance $R = 5.0\ \Omega$.

1. Determine the output voltage (V_o) and load current magnitude (I_o).
2. Determine the average inductor current (I_L), the ripple current (Δi_L), and the peak and minimum inductor currents ($I_{\max}$ and $I_{\min}$).
3. Verify whether the converter operates in Continuous Conduction Mode (CCM).
4. Calculate the peak-to-peak output voltage ripple (ΔV_o) and percentage ripple.

Solution:

- **Given Data:** $V_s = 24.0$ V, $D = 0.40$, $f = 100$ kHz $\implies T = 10\ \mu$s, $L = 20\ \mu$H, $C = 80\ \mu$F, $R = 5.0\ \Omega$.

1. Output Voltage and Load Current:

$$V_o = -V_s\left(\frac{D}{1-D}\right) = -24.0 \times \left(\frac{0.40}{0.60}\right) = \mathbf{-16.0}\ \text{V} \implies |V_o| = 16.0\ \text{V}$$

$$I_o = \frac{|V_o|}{R} = \frac{16.0\ \text{V}}{5.0\ \Omega} = \mathbf{3.20}\ \text{A}$$

2. Inductor Current Analysis:

$$I_L = \frac{I_o}{1-D} = \frac{3.20\ \text{A}}{0.60} = \mathbf{5.333}\ \text{A}$$

During switch ON-interval ($DT = 4\ \mu$s):

$$\Delta i_L = \frac{V_s DT}{L} = \frac{24.0 \times 0.40 \times (10 \times 10^{-6})}{20 \times 10^{-6}} = \frac{9.6}{2.0} = \mathbf{4.80}\ \text{A}$$

$$I_{\max} = I_L + \frac{\Delta i_L}{2} = 5.333 + 2.40 = \mathbf{7.733}\ \text{A}, \qquad I_{\min} = 5.333 - 2.40 = \mathbf{2.933}\ \text{A}$$

3. CCM Verification:
Since $I_{\min} = 2.933\ \text{A} > 0$, the inductor current is always positive. **CCM operation is verified.**
4. Output Voltage Ripple:

$$\Delta V_o = \frac{I_o D}{Cf} = \frac{3.20 \times 0.40}{(80 \times 10^{-6}) \times (100 \times 10^3)} = \frac{1.28}{8.0} = \mathbf{0.160}\ \text{V} = \mathbf{160}\ \text{mV}$$

$$\%\ \text{Ripple} = \frac{\Delta V_o}{|V_o|} \times 100 = \frac{0.160}{16.0} \times 100 = \mathbf{1.00}\%$$

Summary of Results (Question 14):
$V_o = -16.0$ V | $I_o = 3.20$ A | $I_L = 5.333$ A | $\Delta i_L = 4.80$ A | $I_{\max} = 7.733$ A | $\Delta V_o = 160$ mV (1.0%)

Question 15: Buck-Boost Converter Design — Component Sizing & Device Ratings

Textbook Reference: Daniel W. Hart, Chapter 6 / Ned Mohan, Section 7-5.

Problem Statement:
Design an inverting Buck-Boost regulator to deliver $V_o = -18.0$ V to a load of $P_o = 36.0$ W ($R = 9.0\ \Omega$) from an input supply of $V_s = 12.0$ V. The specifications are: $f = 50.0$ kHz, voltage ripple $\Delta V_o/|V_o| \leq 1.0\%$, and inductor current ripple $\Delta i_L \leq 25\%$ of I_L.

1. Determine the required duty cycle D and load resistance R.
2. Calculate the critical inductance ($L_{\min}$) for CCM, and size the filter inductance (L).
3. Size the filter capacitance (C).
4. Determine the peak voltage and peak current ratings required for the active switch and diode.

Solution:

- **Given Data:** $V_s = 12.0$ V, $|V_o| = 18.0$ V, $P_o = 36.0$ W, $f = 50$ kHz $\implies T = 20\ \mu$s, $\Delta V_o/|V_o| \leq 0.01$.

1. Duty Cycle and Load Current:

$$\frac{D}{1-D} = \frac{|V_o|}{V_s} = \frac{18.0}{12.0} = 1.50 \implies D = 1.50(1-D) \implies 2.50D = 1.50 \implies \mathbf{D = 0.60}$$

$$I_o = \frac{P_o}{|V_o|} = \frac{36.0\ \text{W}}{18.0\ \text{V}} = \mathbf{2.00}\ \text{A}, \qquad R = \frac{|V_o|}{I_o} = \frac{18.0}{2.00} = \mathbf{9.00\ \Omega}$$

$$I_L = \frac{I_o}{1-D} = \frac{2.00}{0.40} = \mathbf{5.00}\ \text{A}$$

2. Inductor Sizing:
Critical boundary inductance:

$$L_{\min} = \frac{(1-D)^2 R}{2f} = \frac{(0.40)^2 \times 9.00}{2 \times 50000} = \frac{0.16 \times 9.00}{100000} = \mathbf{14.40}\ \mu\text{H}$$

Design requirement: $\Delta i_L \leq 25\% \times I_L = 0.25 \times 5.0\ \text{A} = 1.25\ \text{A}$:

$$L = \frac{V_s D}{f \Delta i_L} = \frac{12.0 \times 0.60}{(50 \times 10^3) \times 1.25} = \frac{7.20}{62500} = \mathbf{115.20}\ \mu\text{H} \quad (\gg L_{\min})$$

3. Filter Capacitance Sizing:

$$C \geq \frac{D}{Rf(\Delta V_o/|V_o|)} = \frac{0.60}{9.00 \times (50 \times 10^3) \times 0.01} = \frac{0.60}{4500} = \mathbf{133.33}\ \mu\text{F}$$

Select standard commercial rating: 150 μ**F** or 220 μ**F**.
4. Power Semiconductor Ratings:
In a Buck-Boost converter, when the switch turns off, the switch experiences a voltage equal to $V_s + |V_o|$:

- **Switch (MOSFET):**
  - Peak reverse voltage: $V_{\text{switch, peak}} = V_s + |V_o| = 12.0\ \text{V} + 18.0\ \text{V} = \mathbf{30.0}\ \text{V}$ (select ≥ 60 V rated).
  - Peak current: $I_{\text{switch, peak}} = I_L + \frac{\Delta i_L}{2} = 5.00 + 0.625 = \mathbf{5.625}\ \text{A}$.
  - RMS current: $I_{\text{switch, rms}} \approx I_L\sqrt{D} = 5.00 \times \sqrt{0.60} = \mathbf{3.873}\ \text{A}$.
- **Diode:**
  - Peak reverse voltage: $V_{\text{diode, peak}} = V_s + |V_o| = \mathbf{30.0}\ \text{V}$.
  - Peak forward current: $I_{\text{diode, peak}} = \mathbf{5.625}\ \text{A}$.
  - Average forward current: $I_{\text{diode, avg}} = I_o = \mathbf{2.00}\ \text{A}$.

Summary of Results (Question 15):
$D = 0.60$ | $L_{\min} = 14.4\ \mu$H | Designed $L = 115.2\ \mu$H | $C \geq 133.33\ \mu$F | $V_{\text{device}} \geq 30$ V, $I_{\text{peak}} = 5.625$ A